



Our Mission - The Impact of Light on Health

Our work centers on the belief that the most influential environmental factor affecting health is both visible and non-visible light. While diet, exercise, toxins, relationships and many other factors certainly impact our well-being, we maintain that achieving optimal health requires addressing one's visible and non-visible light environment as the foundational step as seen through human history.

As we have shifted to indoor living, we have become deficient in full-spectrum sunlight while being increasingly exposed to unbalanced frequencies of artificial light and the pulsing wireless frequencies of an alternating current (AC) power grid which is rife with dirty electricity and magnetic fields. The resulting environment is an entirely alien ecosystem for human biology.

Artificial visible and non-visible wavelengths of light have exploded over the past 30 years and are severely detrimental hazards for human health and longevity. As we gain a deeper understanding of how light interacts with biology at the quantum level, we learn that the human body is a hydrated, carbon photoelectrical system that has used a hydrogen burning engine powered by full spectrum sunlight throughout all human history. The human living system stores this captured energy or charge within the water network and melanin inside the body as DC electricity. The human body runs on a net negative charge and requires a constant input of electrons in order to function optimally. When this connection to nature is severed, our net negative charge drops and we begin experiencing disease. We've come to understand that most diseases are caused by a lack of energy within the body rather than nuclear DNA mutations. Examining the human body through a biophysical perspective is essential for understanding how our environment influences the health or disease of our epigenome, and for unraveling the intricate workings of light's impact on our human living system.

Life on Earth owes its existence to the sun. Sunlight serves as the primary source of energy and information that fuels the growth and flourishing of all living organisms. This phenomenon is commonly recognized in the plant kingdom as photosynthesis. Interestingly, animals and plants alike demonstrate a profound reliance on the 24-hour light and dark cycle, each employing their unique adaptation of photosynthesis, often referred to as "animal photosynthesis."

Light plays a pivotal role in activating both the brain and the body. It permeates the body through the eyes and skin, and the mere presence of a single photon of light is enough to activate numerous processes across the entire brain. This luminous signal sets in motion the hypothalamus, the control center responsible for overseeing vital bodily functions, including the autonomic nervous system, endocrine system, and the pituitary gland, often referred to as the body's master regulator. Furthermore, the hypothalamus takes charge of our body's biological clock. The significance of natural sunlight exposure and the application of light therapy for the well being of the body has been a subject of research and practice since the late 19th century. However, many of these principles have fallen to the wayside in today's approach to health.

We believe a healthy human living under full spectrum sunlight and eating food which contains a matching light signature is a system well-designed by nature to support longevity and vitality.

Consequently, when we deviate from this practice, choosing to live indoors under artificial light sources or consuming out-of-season foods that contain conflicting energy and information signals at inappropriate times of the day, we significantly increase our vulnerability to health risks.

Over the last four decades, we've witnessed significant shifts in the way we live. Perhaps the most underappreciated change which we have selected for is multigenerational indoor living, 24-hour cities and night-shift working. In 2023, the average person in the USA lives 22 hours of their day indoors. The significance of this from a human health perspective can only be appreciated when we realize that for all of human history, we evolved outdoors connected to nature 24/7. Nothing on earth would exist if there was no sunlight. Everything on earth is subject to gravity which pulls us towards the earth's surface as a baseline for living. We are designed to breathe freshly created oxygen and nitrogen created by the plant kingdom around us. We drink clean fresh spring water provided by our abundant water sources. And we eat wild organic animals, fruit, and vegetables to sustain us. This is how we lived for tens of thousands of years and has formed the basis of our health status and both biologic genomes, nuclear DNA and mitochondrial DNA.

When we began our initial migration to indoor living many centuries ago, we lit the spaces with fire light, kerosine lamps and more recently incandescent bulbs. This meant that the 51% of red and infrared light emitted by our sun was still preserved to an extent when we were indoors. We also initially used natural building materials like stone which held the infrared light and released it as heat in the home for several hours after sunset and kept good airflow and allowed natural light in through semi-open spaces in the walls. Fast forwarding to just the last few decades, we have banned incandescent bulbs, are told fire is a hazard and rarely use fire indoors, used double glazed tinted windows, creating an alien spectrum of sunlight entering the home (especially blocking the red and infrared), and insulate our modern concrete, brick or gyprock building materials with foam insulation to avoid nature's seasonal variation which our biology is designed to sense so it can adapt optimally. To add to this, we wear rubber soled shoes, are told to wear facemasks, are told to be afraid of the sun, to put on sunglasses and lather ourselves in sunscreen, all resulting in a numbness to nature's energy and information signals. Since we've migrated indoors, we have not only cut ourselves off from natural light (especially infrared light), the earth's electric and magnetic fields or breathing freshly created air, but we have also introduced artificial fluorescent and LED light sources, artificial wireless communication and entertainment technology and created artificial microenvironments with constant temperature void of seasonality. This shift has played a pivotal role in the upsurge of neolithic diseases, impacting a rapidly growing number of adults and children not only in the United States, but also global populations. Obesity, Diabetes, Autism, Parkinson's Disease (PD), Cancer, Arthritis, insomnia, Chronic or Adrenal Fatigue, Thyroid disease, Cardiovascular Disease, and Mental Health issues like ADHD are exploding in prevalence – in fact, you'd be hard pressed to find a single person who does not know someone or have themselves one of these once uncommon diseases/conditions.

Artificial Visible Light: We experience exposure to artificial visible light most commonly from our technology screens, indoor/outdoor LED or fluorescent lighting, or in our commutes via car, bus, train or plane. All of these lights contain fragmented spikes of blue and green wavelength light. These spikes disrupt our nervous system, endocrine system, and immune systems via its action on our eyes and skin. These isolated blue and green spikes have a primary destructive effect when they are not balanced with the remainder of the full spectrum (purple, cyan, yellow, orange, red and infrared) it destroys visual and non-visual photoreceptors by freeing the covalently bound vitamin A attached to all OPSIN protein photoreceptors, including melanopsin our circadian photoentrainment photoreceptor in our eyes and skin, to not only result in the creation of an retinal aldehyde free radical destroying the ability for that photoreceptor to capture light energy and information signals effectively dysregulating the circadian sense in that cell, but also lowering vitamin D levels even when in direct UV sunlight. Vitamin A and Vitamin D are entangled due to their tight connection in the neuroectoderm during embryological development. This Vitamin A derivative retinal aldehyde release from the circadian photoreceptor melanopsin also destroys melatonin in the blood, leading to all artificially lit homes occupants to become melatonin deficient and thus supplemental melatonin junkies to get a solid night's sleep and regenerate. In addition to this major effect, exposure to this artificial light reduces the charge in our blood (lowers the zeta potential lowering oxygen carrying capacity, adversely impacting circulatory flow and blood pressure), damages our mitochondrial engines

(water and energy production are disrupted), damages lipids like cholesterol and photoreceptive vitamins like B12, destroys peripheral and central circadian time perception turning on/off genes signaling what we call disease, creating confusion within the body otherwise known as inflammation and a myriad of other harmful oxidative cascades. Our days are too dark and our nights are too bright - confusing our biology - demonstrating the link between mitochondrial circadian rhythm disruption and all common diseases today.

Artificial Non-Visible Light: Our exposure to non-visible artificial light is divided into 3 broad categories, radiofrequency radiation (RF), low frequency electric fields (LFE), and low frequency magnetic fields (LFM). RF radiation exposure comes from cell phones, Wi-Fi, Bluetooth, 4G & 5G, Smart Electrical Meters, Virtual Reality Sets, Game Consoles, Wireless baby monitors, RFID chips, Starlink, GPS, Radio, Microwaves, and Satellite communication to name a few. LFE exposure primarily comes from the copper wiring of the AC power grid (60Hz in the USA/Mexico and 50Hz in Europe/Australia/Asia), car electronics, dirty electricity from faulty electrical wiring or improper electrical grounding, and technological devices and appliances like computers and kitchen appliances. LFM radiation can be found mostly from local distribution power lines, solar inverters, battery charging stations, pool pumps, car engines, whitegoods like refrigerators, freezers and washing machines, or other larger mechanical appliances. These electric, magnetic and electromagnetic fields are alien to our biology and impact our biology in several ways – these include – the retinol aldehyde free radical creation, lowered vitamin D and oxidative destructive cascade damaging photoreceptors and lowering melatonin as described above in the artificial visible light section, activating voltage gated calcium channels (VGCCs) resulting in excessive intracellular calcium levels restructuring cell water and destroying biochemical signaling, resulting in oxidative stress (the process of a loss of electrons and thus a loss of electric charge in the body), creation of peroxynitrite free radicals due to excessive superoxide free radical creation damaging cellular architecture and raising inflammation, degradation of water molecules leading to dehydration, buildup of misfolded proteins (linked to neurodegenerative and neurodevelopmental conditions study), increasing gut, thyroid and brain barrier permeability, destruction of sperm count and motility, raising blood glucose independent of food and dysregulating insulin (diabetes alert!) and radiofrequency (RF) radiation has been linked to brain cancer, including glioblastoma multiforme (GBM). (Interphone Study, Hardell Group Studies, CERENAT Study, REFLEX Project, Bioinitiative Report, Study.

Recent research has revealed the importance of visible and non-visible light on human health.

Presently, the biological influence of light on health and disease remains a relatively overlooked aspect in mainstream medicine. However, we hold the view that it's not merely a contributing factor to the rise in neolithic diseases; rather, it represents the principal force behind the escalating prevalence of illness.

In Jie Shen and John Tower's 2019 study titled 'Effects of light on aging and longevity', they state: "Light exposure is an environmental factor that significantly impacts aging and longevity. Light intensity, spectral components, and duration of exposure, are all important factors to consider... Light conditions should be carefully controlled to avoid potential conflicts between different studies."

"Human adaptation to ambient light has two mechanisms: image forming (IF) for vision, and non-image forming (NIF). NIF functions are mediated by retinal ganglion cells (RGCs) expressing the photopigment melanopsin, and the action spectrum of melanopsin photopigment peaks around 480nm blue light [Hatori, Gronfier et al. 2017]"
Effect of Light on Aging + Longevity SCIENCE DIRECT

From these sources and many others, we learn that visible and invisible light, dark and temperature variation are the key controllers of our circadian rhythm. In fact, in a mouse model, it was shown by Dr. John Hogenesch that ~55% of the mouse genome is controlled by circadian clock timing. Meaning that the majority of the proteins synthesized are done so when given a particular light, dark, or temperature signal to do so. "...we continue to

sample additional organs, we predict ~10,901 mouse protein-coding genes (55% of the background set) will show circadian oscillations somewhere in the body." PNAS.ORG Link

The Impact on Children

A specific concern is the increasing incidence of brain diseases in children, who are exceptionally vulnerable to the adverse effects of artificial light and invisible non-native electric and magnetic fields present in modern life. Children's thinner corneas, lenses, and retinas render them highly sensitive to the visible and non-visible light in their surroundings. The primary sources of this exposure include LEDs in home lighting, technology screens, wireless devices, solar inverters, smart meters, home appliances, baby monitors, and smart toys. Children's eyes have larger pupils and more transparent lenses than adults, and light streams into them more freely. Transmission of blue light through a 9-year-old's eye is 1.2-times higher than that of an adult. Light before bedtime, even at low intensities, results in robust and sustained melatonin suppression. In studies of minimal light exposure before bed vs. light exposure— melatonin was suppressed anywhere from 70% to 99% after light exposure. (Ref, Ref, Ref, Ref) Mental health conditions, including depression, are biochemically represented by dysfunctional protein construction within the brain. Normally, these proteins are synthesized through specific frequencies within sunlight interacting with aromatic amino acids such as tyrosine and tryptophan in our eyes and skin. These amino acids are crucial for the creation of vital neurotransmitters like dopamine, serotonin, and melatonin, as well as other essential compounds including thyroid hormones, nicotinamide adenine dinucleotide (NAD), and various hormones produced from the cleaving of proopiomelanocortin (POMC) in our eye and skin. It should be blatantly obvious to anyone paying attention that technology companies are aware of this impact. Companies like Google, Facebook, and Apple have patents on blue light-emitting screens, which influence mental health. With children now addicted to technology, it is no surprise that their mental abilities and moods are being adversely affected. With kids now addicted to technology is no shock how children's mental ability, moods, and Is this the reason why record numbers of kids are now being placed on melatonin? YES.

Measurements reveal that the LEDs and devices we commonly employ indoors emit unnatural spikes of harmful light. Given the pivotal role of light in regulating the endocrine and nervous systems from the eye to the brain, children exposed to these visible and non-visible fields face a heightened risk of adverse health outcomes. Neurodegenerative diseases and conditions affecting brain development, addiction, behavioral challenges, diabetes, and even cancer, have surged in children over the past three decades.

The Impact on Sleep

Why has achieving a good night's sleep become such a challenge for many in today's artificially lit and device-oriented world?

Sleep-related issues and disorders have, unfortunately, become pervasive across all age groups, from children and teenagers to adults. These issues have disrupted sleep patterns and, somewhat alarmingly, have become normalized over the past 15 years. However, sleep is now being recognized as a fundamental pillar of overall health. As this awareness has grown, an increasing number of individuals are turning to us for effective sleep solutions, and in doing so, we've had the privilege of transforming the lives of tens of thousands of health-conscious individuals.

Our expertise in the domain of sleep rests upon decades of sleep science not only confirming the critical significance of high quality sleep, but also shed light on how visible and non-visible light influences sleep hormones like melatonin, cortisol, leptin, and growth hormone. These hormones both exert a substantial impact on and are influenced by the sleep-wake cycle, thus shaping our overall health and well-being.

Sleep is not merely a passive state; it is an active process that warrants respect and priority. Within the confines of a good night's sleep, numerous essential processes unfold. These encompass optimizing body composition,

regenerating tissues, enhancing ketogenic metabolism, and fostering the growth of neuronal networks. In essence, sleep acts as a rejuvenating and transformative force that influences our physical and mental health in profound ways.

Our Mission

Our mission is to raise awareness about the crucial impact of light on human biology. We educate people and organizations about the biophysics of light and its effects on human health through seminars, webinars, retreats, and numerous educational programs.

Our partners include medical facilities, sports organizations, specialized healthcare professionals, professional athletes and business leaders. Our engagement in formal scientific studies is advancing the fields of neuroscience, biophysics, and optics and demonstrating the central role that visible and non-visible light plays in regulating critical biochemical processes.

Our purpose is to revolutionize the healthcare paradigm by explaining the 'why' behind the escalating rates of autism, ADHD, anxiety, depression, Parkinson's, Alzheimer's, sleep disorders, obesity, diabetes, autoimmune conditions, and various other diseases, conditions, and syndromes and what can absolutely be done about them. Together we empower people to embark on a journey of true healing, rather than merely addressing symptoms. Through our many avenues and efforts, we aspire to catalyze transformative changes in lives, fostering a comprehensive approach to wellness and health-span.

Education

We provide educational seminars, workshops, retreats, and programs designed to enhance health and wellness through an understanding of biophysics and its central role in human health. Our comprehensive sessions cover various topics aimed at unlocking the once secrets to optimal health, focusing on the interconnected roles of sleep, light, hydration, food and overall biological energy generation. We empower individuals with the knowledge and tools needed to achieve vibrant health and longevity by addressing the root causes of chronic diseases and leveraging the latest scientific research and technological advancements.

Our "Beyond Illumination Educational Sessions" offer a deep dive into topics such as: Cracking the sleep code for better rest and regeneration, The influence of light on brain health and disease prevention, The role of cellular hydration in health optimization and the biological energy generation beyond food and water.

Additionally, we explore the impact of sunlight on biochemical reactions, strategies for reversing chronic disease without drugs or supplements, and the importance of understanding radiation's effects on our biology. From enhancing brain development in children to optimizing fertility and hormonal health, our programs provide actionable steps to improve your sleep, brain function, gut health, immune system, stress resilience, and overall health span.